

EDA Project

EDA- Project Study Points

- Presentation: 19th-23rd January 2026
- Deadline for submission: 18th January 2026
- Deadline for code and result: 15th January 2026
- Report writing: 16-17-17 Jan

Project Report Structure Criteria and requirements:

- **Sreyleak**: Introduction, Data Preprocessing, EDA(quantitative variables)
- **Sreypov**: EDA(categorical variables), Hypothesis Testing, Conclusion
- **Sreyleap**: Model Building, Results and Evaluation

1. Introduction

- Objective and purpose of the project
- Provided dataset overview, descriptive statistics

2. Data Preprocessing

- Detail step taken to clean and preprocess the data
- Explain handling of missing values and outliers.
- Describe any feature engineering techniques applied.

3. Exploratory Data Analysis (EDA)

Include visualizations (charts, graphs) and descriptive statistics.

Identify key patterns, trends, and anomalies in the data. (Quantitative and Qualitative)

- Univariate analysis
- Bivariate and Multivariate analysis
- Correlation analysis(pearson, spearman)
- outliers detection (boxplot, z-score, IQR)

4. Hypothesis Testing

- Study the relationships between catagorical variable
- Formulate hypotheses based on EDA findings.
- Conduct statistical tests to validate or refute hypotheses.
- PCA, or CA
- chi-square test for two catagorical variables, use CA or PCA, visualizations and interpret

5. Model Building Unsupervised/supervised Tasks:

- Exploring structures or connections of the data according the flow of your analysis.
- Provide details on clustering methods or model training, validation, and testing processes
- Include any choice of parameters of unsupervised tasks or hyperparameter tuning and model

6. Results and Evaluation

- Present the resulting performance (e.g., obtained structures or accuracy, precision, recall, score, RMSE).
- Compare performance across different methods if multiple were used.
- Highlight any insights or conclusions drawn from the implemented methods/models.

7. Conclusion:

- Summarize key findings and their implications.
- Discuss any limitations of the analysis.
- Suggest potential areas for future research or improvement.

8. References:

- Cite all sources of data and any external libraries or tools used.
- Include academic references for any theoretical concepts or methods applied.